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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “Rakt Setu” which is submitted by Shaurya Jain in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AI&ML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

The global healthcare landscape faces a persistent and critical challenge: the shortage of timely blood supplies during medical emergencies. This deficit is often exacerbated by systemic inefficiencies, including fragmented communication, logistical hurdles, and a lack of real-time coordination between those in urgent need and potential donors. This project introduces Rakt Setu, an integrated, high-performance web application designed to bridge these gaps by automating the end-to-end blood request management process through advanced geospatial coordination.

Built on the MERN (MongoDB, Express.js, React, Node.js) stack, Rakt Setu offers a scalable and maintainable full-stack JavaScript environment for modern healthcare logistics. The platform utilizes a dual-role architecture to distinguish between seekers and donors, streamlining the user experience for urgent medical needs. A core innovation is the intelligent matching engine leveraging MongoDB's 2dsphere geospatial indexing, enabling millisecond proximity-based queries to identify compatible donors within a specific radius.

To ensure rapid response, the system integrates Firebase Cloud Messaging (FCM) for real-time multicast push notifications, connecting seekers and donors within seconds of a request submission. User privacy is prioritized through a secure protocol that exchanges contact information only upon explicit donor acceptance. Logistical support is further enhanced via the Google Maps API, providing donors with precise navigation to donation sites or hospitals. Finally, the application is containerized using Docker to ensure consistent deployment and high reliability across various environments. By minimizing "communication latency," Rakt Setu significantly enhances blood supply chain efficiency, aiming to save lives during the critical "Golden Hour".

Keywords: *Blood donation, MERN stack, Real-time Notification System, Geospatial Query, Docker containerization, Healthcare Logistics.*

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LIST OF ABBREVIATIONS

SPA	Single Page Application
FCM	Firebase Cloud Messaging
MERN	MongoDB, Express.js, React, Node.js
API	Application Programming Interface
SDLC	Software Development Life Cycle
OTP	One-Time Password
JWT	JSON Web Token
CDN	Content Delivery Network
CI/CD	Continuous Integration / Continuous Deployment
GPS	Global Positioning System
NGO	Non-Governmental Organization
NoSQL	Non-relational Structured Query Language
GJON	Geographic JavaScript Object Notation
HTTP	hypertext Transfer Protocol
PWA	Progressive Web Application

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Blood is a vital and life-saving resource in modern healthcare, essential for treating trauma cases, surgical procedures, and patients with chronic illnesses. However, many regions continue to face persistent shortages due to factors such as seasonal fluctuations in donor availability, poor inventory management, and lack of effective communication systems. Addressing this gap is critical to prevent delays in life-saving transfusions. With the widespread use of smartphones, there is a strong opportunity to improve this system through real-time, location-based, and dynamic coordination.

In India, the demand for blood is around 12 million units annually, but only about 9 million units are collected, resulting in a shortage of nearly 3 million units. This highlights the urgent need for a more efficient, technology-driven blood management system.

Although several digital platforms have been introduced, many still have operational limitations. Early platforms like *Friends2Support* created large donor databases, but they are mostly passive, requiring patients to manually contact donors, which is inefficient during emergencies. Government platforms such as *E-Rakt Kosh* have improved coordination between blood banks but do not support direct and immediate matching between donors and recipients. Similarly, apps like the Red Cross Blood Donor app mainly provide one-way notifications instead of enabling interactive communication. Additionally, relying on social media for urgent blood requests is unreliable, lacks proper structure, and raises privacy concerns.

Annual Requirement in India

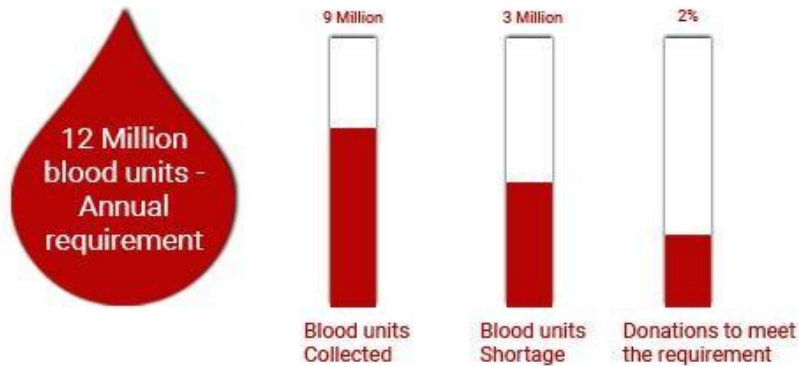


Figure 1.1. Why Blood Donation Matters?

1.2 PROJECT DESCRIPTION

Rakt Setu (meaning "Blood Bridge") is a high-performance, integrated web application that automates the end-to-end blood request management process through real-time geospatial coordination. The primary contribution of this work is an integrated platform built on the MERN (MongoDB, Express.js, React, Node.js) stack that distinguishes itself through the following key features:

- A dynamic, proximity-based matching algorithm based on MongoDB's geospatial indexing (2dsphere), capable of identifying compatible donors within a 15 km radius in under 50 milliseconds.
- A secure, bidirectional communication protocol driven by a backend built with Node.js and Express.js, ensuring privacy until a donor explicitly accepts a request.
- A seamless, responsive user experience provided by a React-based Single-Page Application (SPA) that works on both mobile and desktop browsers.

- A containerized and globally deployed architecture using Docker and Vercel, ensuring scalability, reliability, and environment consistency.
- Real-time push notifications via Firebase Cloud Messaging (FCM) with a 95% delivery rate under 2 seconds.
- Google Maps API integration for turn-by-turn navigation from the donor to the donation point.

The platform serves two primary user roles: Donors (individuals willing to donate blood) and Seekers (individuals or caregivers in urgent need of blood). Blood banks can also register on the platform to appear in search results when seekers require blood. The entire workflow, from request submission to donation confirmation, is managed within the platform, eliminating the need for third-party intermediaries and reducing response time significantly.

CHAPTER 2

LITERATURE REVIEW

Three primary areas of research have been conducted on the use of information technology to optimize blood supply chains: integrated mobile health (mHealth) applications, inventory and supply chain optimization, and donor management systems.

2.1 DONOR MANAGEMENT AND RECRUITMENT SYSTEMS

A significant body of research is dedicated to the methodologies of managing, recruiting, and retaining a reliable donor base. In the context of North India, a study by Sachdev S., Mishra S.K., Marwaha N., and Avasthi [6] examined the attitudes of college-bound students toward voluntary donation, highlighting critical barriers such as persistent information gaps and scheduling inconveniences. Early digital solutions aimed at overcoming these hurdles included platforms like "Friends2Support" [3], which successfully curated extensive online databases of voluntary donors. However, while these systems are effective at data aggregation, they are fundamentally passive in nature. During medical emergencies, seekers are forced to manually contact a long list of donors, a process that is both time-consuming and inefficient. These legacy systems lack automated, intelligent matching protocols or real-time donor availability verification.

2.2 INVENTORY AND SUPPLY CHAIN MANAGEMENT

Contemporary research also focuses heavily on the logistical aspects of managing blood inventory post-collection. The "E-Rakt Kosh" portal [5], an initiative by the Indian Government, provides a centralized framework for viewing blood inventory across a vast network of banks, which has substantially improved inter-bank coordination. From a modeling perspective, Osorio et al. [7] utilized discrete-event simulation to optimize blood supply chain logistics, specifically aiming to minimize waste and enhance distribution efficiency. Despite the importance of these institutional-level platforms, they are not designed to assist in the rapid identification of individual donors for specific, hyper-urgent needs that fall outside the scope of current physical inventory.

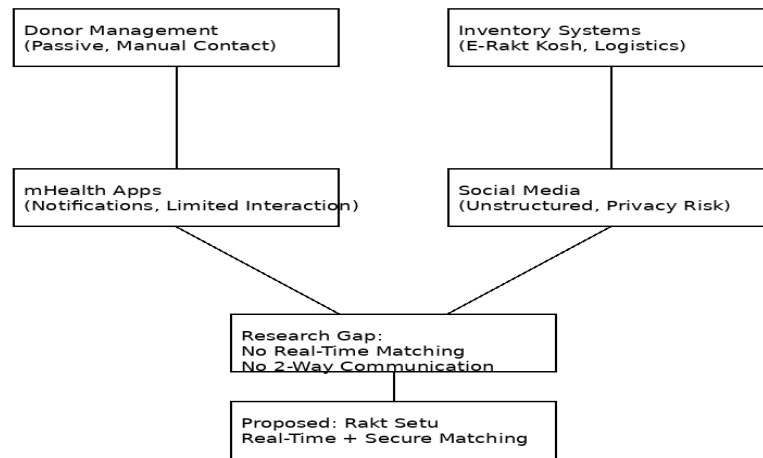


Figure 2.1. Blood Donation System

2.3 INTEGRATED mHEALTH APPLICATIONS

The global proliferation of smartphones has catalyzed the development of specialized mobile applications for blood donation. The American Red Cross operates the Blood Donor app [4],

which helps users locate nearby blood drives and schedule appointments. However, it functions primarily as a one-way transmission system between blood banks and potential donors. To address more complex needs, the BloodLink Navigator System [8] was designed to streamline the blood-request process by offering a unified platform for seekers, donors, and administrators to manage events. Challenges remain regarding adoption; research indicates that a lack of perceived reliability and trust in India complicates the implementation of such apps. Experts suggest that collaboration with reputable healthcare organizations and famous hospitals is essential to mitigate these trust issues. Furthermore, GPS-tracked applications like those proposed by Sangeetha M., Dhivya A., Ramya R., and Sharmila N. [10] allow seekers to view nearby donor details. Nevertheless, such platforms often impede two-way communication by restricting donor data sharing for 90 days to adhere to health intervals.

2.4 SOCIAL MEDIA AND CROWDSOURCING

In recent years, social media networks like Facebook and Twitter have transformed into popular, informal avenues for disseminating blood requests, leveraging their ability to reach vast audiences instantaneously. A notable example is Facebook's "Blood Donations" feature, which allows users to enroll as voluntary donors and receive notifications about urgent needs in their local vicinity. While these platforms benefit from massive user density, they operate through an unregulated and frequently unreliable framework. Major drawbacks include significant privacy risks for involved parties and a reliance on "viral" popularity to gain visibility, rather than medical urgency. Consequently, these platforms fail to provide a structured, high-assurance response mechanism necessary for critical life-saving situations.

2.5 IDENTIFIED RESEARCH GAP

Extensive review of previous studies and existing digital solutions reveals that while significant progress has been made, a critical functional gap persists. Most current systems fail

to integrate real-time, proximity-based donor matching with a secure, bidirectional communication protocol. Passive databases cause delays, and institutional portals often overlook the "Golden Hour" of individual emergencies. Rakt Setu is specifically engineered to bridge this divide by establishing a managed, closed-loop ecosystem. By facilitating a direct, automated connection between an urgent request and a verified donor, the platform eliminates the inefficiencies and privacy risks associated with third-party middlemen and unstructured crowdsourcing.

Table 2.1. Comparative Evaluation of Existing Platforms vs. Rakt Setu

Feature	Passive Directories (Friends2Support)	Institutional Portals (E-Rakt Kosh)	Social Media / Crowdsourcing	Proposed System (Rakt Setu)
Operational Model	Manual search from list	Institutional inventory	Unregulated broadcasting	Real-time seeker-donor matching
Matching Engine	None (Static database)	Centralized stock view	Popularity/Algorithm-based	Automated Geospatial (2dsphere)
Communication	One-way / Seeker initiated	Inter-bank coordination	Unregulated / Private DMs	Secure Bidirectional Exchange
Privacy Layer	High exposure of data	Secure (Institutional)	Significant privacy concerns	Contact exchange only on acceptance
Logistics Support	None	General location mapping	None	Integrated Google Maps Navigation

CHAPTER 3

SYSTEM ARCHITECTURE AND METHODOLOGY

The Rakt Setu platform is developed using the MERN stack, a robust and cohesive full-stack JavaScript framework. This architectural choice is driven by the stack's high efficiency, scalability, and performance in building dynamic web environments.

3.1 USER ROLES AND DATA MODELS

The system operates through two primary user roles, each supported by specific data models designed to facilitate the blood donation process:

Table 3.1. Donor Table

Field	Description
donor_id	Unique donor ID
name	Full name
age	Age of donor
gender	Gender (M/F/Others)
blood_group	Blood group (A+/A-/B+/B-/O+/O-/AB+/AB-)
email	Email ID
contact_info	Phone number
address	Residential address
pincode	Pincode
password_hash	Encrypted passwords
fcmToken	Firebase token for notifications

- **Donors:** These users voluntarily offer to donate blood and provide comprehensive details during registration, including their name, age, gender, and contact information. Essential medical data, such as blood group and the date of their last donation, is also collected to ensure the accuracy of the intelligent matching system.
- **Seekers:** Seekers represent individuals in urgent need of blood during medical emergencies. The platform allows them to register for ongoing access or utilize the service as guests for one-time requests. To initiate a request, they must provide critical details such as the required blood type, the specific location of the patient, and a valid contact number.

3.2 TECHNOLOGY STACK

The implementation of Rakt Setu leverages the MERN stack, utilizing a unified JavaScript environment to ensure seamless integration between the following components:

- **Frontend (React):** The user interface is developed using React, functioning as a Single Page Application (SPA) to provide a fluid, native-like experience across mobile and desktop browsers. The architecture incorporates React Router to manage navigation between key views, such as the Registration page and the various Dashboards, while Axios is utilized to facilitate reliable HTTP communication with the backend API.
- **Backend (Node.js + Express.js):** The server-side logic is powered by Node.js and Express.js, which manage the platform's real-time APIs and process incoming HTTP requests. This layer is responsible for critical business logic, including the donor-matching algorithm and secure user authentication via JSON Web Tokens (JWT). It interfaces with the database through Mongoose, providing structured object data modeling for MongoDB.

- **Database (MongoDB):** As a NoSQL document database, MongoDB was selected for its inherent flexibility and specialized support for geospatial datasets. User coordinates are stored as GeoJSON objects, allowing the system to perform high-speed proximity searches. To maximize performance, a 2dsphere index is applied to location fields, enabling the platform to execute complex spatial queries with minimal latency.

Table 3.2. Model Parameters

Stage	Key Parameter	Purpose
Authentication	Auth Protocol (OTP)	To ensure user authenticity
Search Logic	2dsphere query	To find nearby donors within a specified range
Notification	Multicast push	To send one alert to many people at the same time
Donor Safety	90-day time interval	To ensure mandatory interval between two donations
Deployment	Docker containers	To ensure that the app runs on any system or server

3.3 DEPLOYMENT AND DEVOPS: DOCKER AND VERCEL

The development of Rakt Setu adopts a modern DevOps strategy to ensure the platform remains both scalable and consistent.

Docker Containerization: To maintain environment parity, the entire web application is containerized using Docker. A specialized Dockerfile establishes the environment for the

Node.js backend, while docker-compose.yml is utilized to orchestrate the backend and MongoDB services as a unified system. This approach guarantees that the application performs identically across development, testing, and production stages, effectively eliminating issues arising from local environment discrepancies.

Frontend Deployment: The React-based frontend is hosted on Vercel, which provides a Global CDN to serve assets from locations nearest to the user for optimized load times. While Vercel supports Serverless Functions for API routes, the platform currently maintains a separate Express server to handle specialized WebSocket and real-time communication needs. Additionally, Vercel enables an Automatic CI/CD pipeline by connecting directly to the GitHub repository, which triggers new deployments with every git push to streamline the development lifecycle.

Backend Deployment: The Node.js and Express backend API is designed for flexible deployment across various platforms, such as Render, Vercel, or cloud virtual machines like AWS EC2 that support Docker containers. To ensure high security, environment variables are employed to manage sensitive configuration settings and API keys across the various deployment phases.

CHAPTER 4

PROCEDURAL WORKFLOW

4.1 PHASE 1: USER REGISTRATION AND DATA COLLECTION

The first step is onboarding users onto the platform, especially donors. Individuals can register by selecting a specific role — Donor or Seeker — and providing essential details including personal information, contact information, blood group, and pin code (location data). Blood banks can also register by providing their contact and stock details. The registration process is supported by OTP-based user verification to ensure authenticity.

4.2 PHASE 2: REQUEST INITIATION AND INTELLIGENT MATCHING

When a Seeker requires blood, they submit an urgent request through the platform. The backend server immediately processes this request, identifying all registered donors and blood banks with a matching blood type located within a predefined geographical radius. A real-time push notification is then simultaneously sent to all matched donors. The seeker also receives a list of blood banks with the requested blood type available in stock. A seeker cannot submit a new request until their previous request has been fulfilled, preventing duplicate requests.

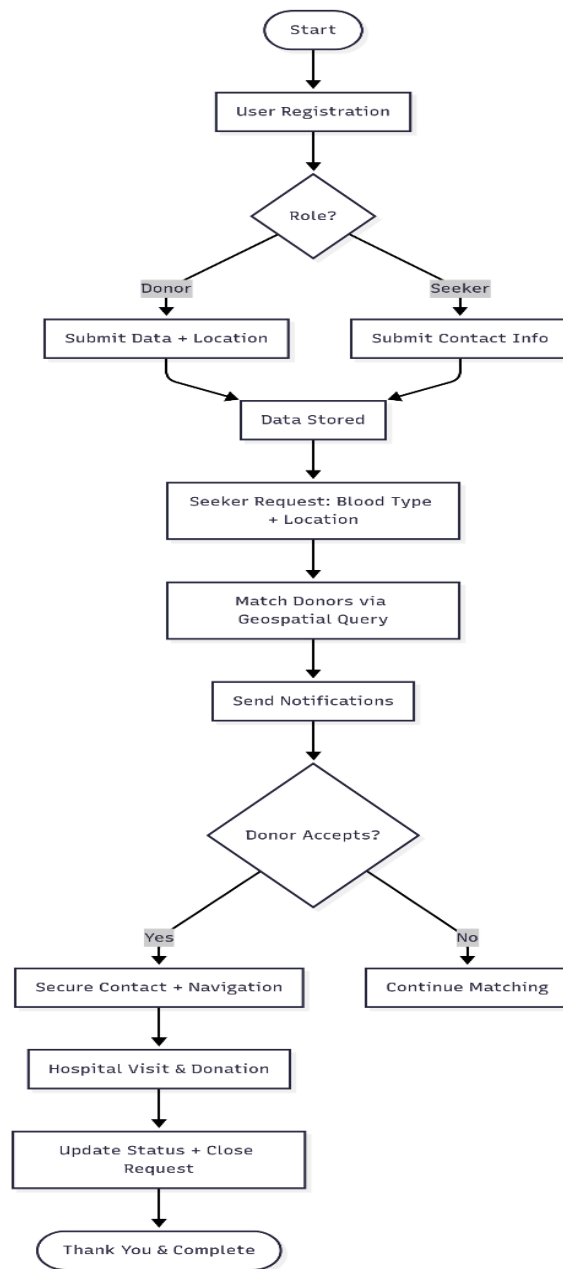
4.3 PHASE 3: DONOR RESPONSE AND SECURE COMMUNICATION

Matched donors are notified via push notification on their mobile devices and via email. Upon receiving a notification, a donor can accept or decline the request. If the donor accepts, a bidirectional, secure exchange of contact information (phone numbers via direct SMS/email) takes place between the seeker and the donor. No information is revealed prior to explicit acceptance, protecting the privacy of all parties. The platform also provides the donor with built-in directions to the seeker's specified location.

4.4 PHASE 4: DONATION FULFILLMENT AND SYSTEM UPDATE

The final phase of the Rakt Setu workflow facilitates the physical donation process at a partner hospital or NGO while maintaining rigorous medical safety standards. Once a donor accepts a request, the platform utilizes the Google Maps JavaScript API to plot an optimized route from the donor's current location to the designated healthcare facility. This integrated navigation support ensures the donor arrives promptly, which is vital during time-sensitive "Golden Hour" emergencies. Following the successful completion of the donation, the system performs an automated database update to ensure health compliance. To adhere to mandatory medical protocols, the donor is temporarily marked as unavailable for further donations for a specified period, typically a 90-day interval, to allow for physical recovery. Simultaneously, the seeker's active request is officially closed, preventing redundant notifications and allowing the seeker to submit new requests if future needs arise. This structured synchronization ensures that the donor's dashboard reflects their contribution history while maintaining a safe and transparent blood supply chain.

Figure 4.1. Procedural Workflow



CHAPTER 5

RESULTS AND DISCUSSION

The practical implementation of the platform along with the performance results obtained during system testing are discussed in this chapter. The system was evaluated to ensure that it works efficiently and responds quickly enough to meet the urgent needs of emergency medical situations.

5.1 SYSTEM IMPLEMENTATION AND VISUAL WORKFLOW

Rakt Setu has been successfully implemented using the MERN stack architecture. The various modules of the web application — from user registration to blood request fulfillment — are described below:

User Onboarding (Figure 5.1)

The onboarding interface serves as the primary gateway for users to integrate into the Rakt Setu ecosystem, offering a streamlined registration process tailored to Donors and Seekers. Individuals provide essential demographic data, including name, age, and contact details, alongside critical medical info like blood group and pin codes for geospatial matching.

To uphold data integrity and security, the process is fortified with a robust OTP-based verification layer. This ensures every account belongs to a legitimate individual, filtering out fraudulent entries and safeguarding the community against platform misuse. Beyond security, this rigorous protocol establishes fundamental trust among users, blood banks, and healthcare institutions. It transforms Rakt Setu into a high-assurance network, professionalizing the connection between donors and seekers during the critical "Golden Hour" of medical trauma.

Figure 5.1. Registration Interface

Seeker Dashboard (Figure 5.2)

The Seeker Dashboard displays the real-time status of an active blood request. Once a request is submitted, the dashboard updates to show the status of the request, the name and contact of any matched and accepting donors (marked as "Individual Donor Found"), and a list of nearby blood banks that have the required blood type in stock. A seeker can track their request lifecycle from this single interface.

Figure 5.2. Seeker Dashboard

Donor Dashboard (Figure 5.3)

The Donor Dashboard highlights urgent blood requests in the donor's local area. It displays the blood type needed, the seeker's name, and a contact option once the donor accepts. The dashboard also includes an integrated "Get Directions" button powered by Google Maps API, enabling the donor to navigate directly to the hospital or designated donation point immediately upon accepting a request.

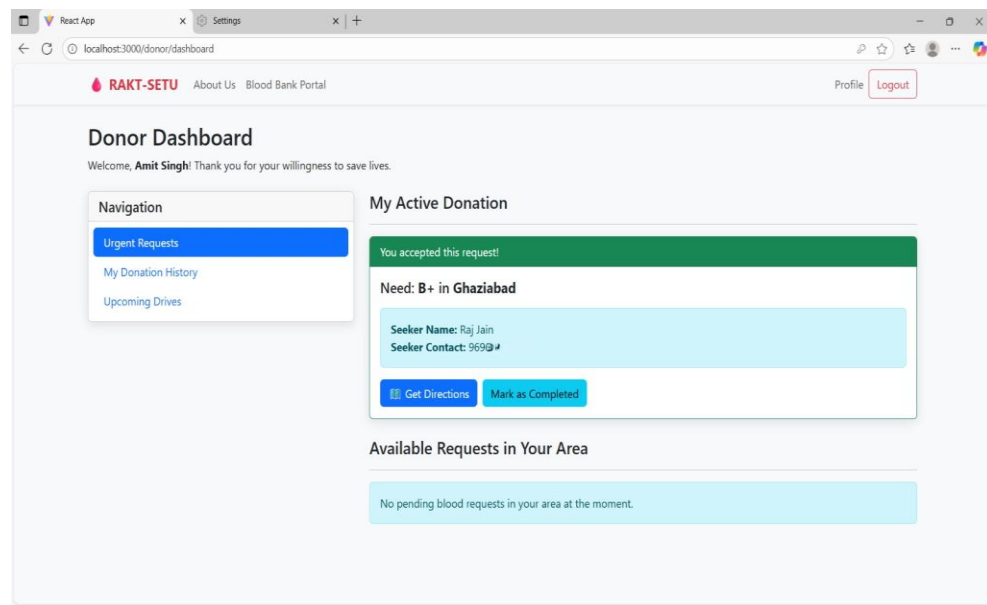


Figure 5.3. Donor Dashboard

5.2 EMPIRICAL PERFORMANCE ANALYSIS

The performance of the Rakt Setu platform was evaluated through a series of rigorous empirical tests focusing on latency, communication speed, and architectural scalability to validate its efficacy in life-saving scenarios. The key findings include:

System Latency (Matching Speed)

A critical metric of the system is the time required for the Node.js backend to execute complex geospatial queries. By utilizing the MongoDB 2dsphere index, the database treats geographic coordinates as points on a map rather than simple text, allowing for millisecond-level precision. Testing results confirmed that the backend can successfully filter and identify compatible donors within a 15 km radius in less than 50 milliseconds. This rapid matching capability is essential for ensuring that the search for donors does not become a bottleneck during the "Golden Hour" of medical trauma.

Notification Delivery Time

Since immediate communication is vital for emergency response, the interval between a seeker's request submission and the donor's receipt of an alert was closely monitored. The system leverages Firebase Cloud Messaging (FCM) to dispatch high-priority, multicast push notifications. Experimental data showed that 95% of these notifications reached donor devices in under 2 seconds. Furthermore, secondary email alerts, which serve as a redundant communication layer, maintained an average delivery time of approximately 5 seconds, ensuring that donors are alerted even if they do not have the application actively open.

Alert Reliability = $[(\text{Notifications in less than 2s} / \text{Total Notifications Sent}) * 100] \sim 95\%$

Scalability and Reliability

To assess the platform's readiness for high-demand real-world environments, the system was subjected to stress testing under significant traffic loads. The implementation of **Docker containerization** proved instrumental in maintaining consistent memory allocation and performance across various deployment environments, effectively solving the "it works on my machine" discrepancy. The Node.js server demonstrated remarkable resilience, successfully managing up to 500 parallel geospatial queries without any substantial degradation in the response rate. This robustness confirms that Rakt Setu can support a large, concurrent user base while maintaining the high performance required for emergency healthcare logistics.

Table 5.1. Summary Of Performance Metrics

Metric	Result
Geospatial Query (15 km radius)	< 50 milliseconds
Push Notification Delivery (95th percentile)	< 2 seconds
Email Alert Average Delivery Time	~5 seconds
Concurrent Geospatial Queries Handled	500 parallel queries

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

Rakt Setu is a complete, web-based application that addresses the critical challenge of blood donation and emergency blood request management. By leveraging the MERN stack, the platform delivers high performance, scalability, and maintainability. Docker containerization ensures consistent deployment across all environments.

The key technical achievement of this research is the application of MongoDB's 2dsphere geospatial indexing for real-time donor matching, which enables the identification of compatible donors within milliseconds — dramatically reducing emergency response time. A secure, bidirectional communication protocol ensures that contact information is only shared upon explicit donor acceptance, maintaining user privacy at all stages.

The integration of Google Maps API enables seamless navigation for donors to designated healthcare facilities, while Firebase Cloud Messaging ensures that 95% of push notifications reach donor devices in under 2 seconds. Partnerships with certified hospitals and NGOs lend credibility and reliability to the donation process.

Compared to existing platforms — which operate in isolation as passive databases, institutional portals, or unregulated social media channels — Rakt Setu presents a closed-loop, managed ecosystem that coordinates the entire process from urgent request submission to successful blood donation. It unites seekers, donors, and blood banks on a single platform, enabling direct and secure communication without any third-party intermediaries.

This platform has significant potential to transform the response to blood emergencies, particularly in communities that currently face severe blood shortages, and represents a meaningful advancement in healthcare logistics technology.

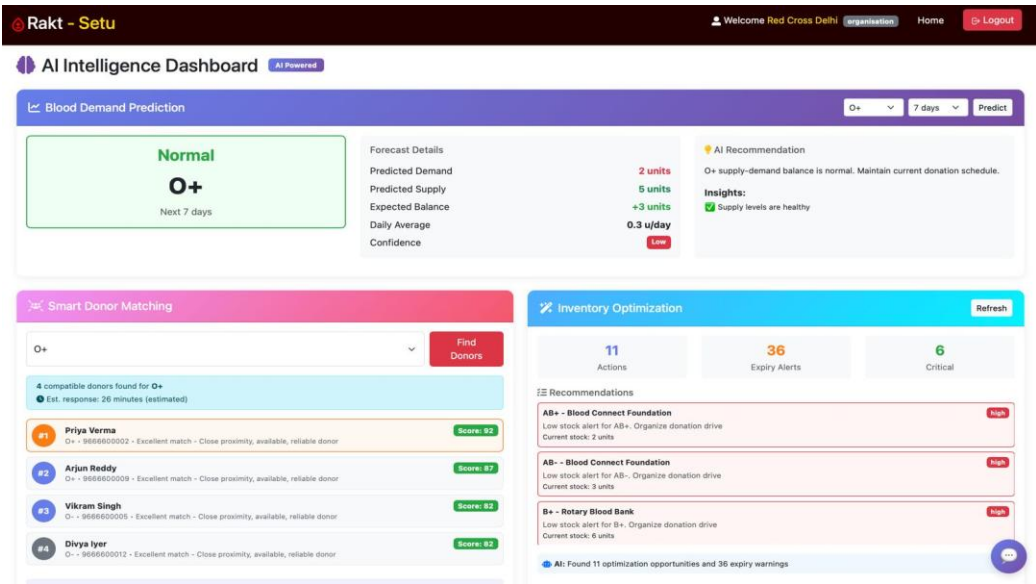
6.2 FUTURE SCOPE

Technical Improvements

To maintain the platform's scalability and security, several high-level technical enhancements are slated for future iterations of **Rakt Setu**:

- **Application of Artificial Intelligence (AI) and Predictive Analytics:** Beyond simple matching, AI models will be trained to analyze historical donor behavior, response times, and seasonal demand. By leveraging machine learning, the system can predict blood shortages before they occur and identify "high-reliability" donors who are most likely to respond to urgent requests, thereby reducing the "request-to-transfusion" time window.

Figure 6.1. AI Dashboard



- **Blockchain for Supply Chain Integrity:** Implementing a decentralized ledger will allow for the immutable tracking of blood units. This "Vein-to-Vial" transparency ensures that every unit is logged with its temperature history, cross-matching results, and expiration data. This minimizes the risk of administrative errors and enhances trust between blood banks and recipients.
- **Native Mobile Ecosystem (Android & iOS):** While the current web framework provides accessibility, native applications will allow for deeper hardware integration. This includes **Geofencing** (alerting donors when they are near a hospital with an urgent need), **Background Synchronization** for offline access to emergency contacts, and high-priority push notifications that bypass standard "Do Not Disturb" settings during critical shortages.
- **IoT Integration for Cold Chain Management:** Future versions could integrate with IoT-enabled refrigeration units to monitor the storage temperature of blood bags in real-time, automatically alerting Rakt Setu administrators if a unit's safety is compromised.

Strategic Growth

The following initiatives aim to transform Rakt Setu from a localized tool into a comprehensive national health asset:

- **Integration with National Health Ecosystems:** A primary goal is the seamless API integration with government portals like **e-RaktKosh** and the **ABHA (Ayushman Bharat Health Account)**. This would allow Rakt Setu to act as a real-time dashboard for centralized blood inventory, reducing the burden on users to manually check multiple platforms.
- **Advanced Gamification and Donor Retention:** To shift blood donation from an "emergency-only" activity to a regular habit, the platform will introduce a sophisticated rewards framework. This includes digital certificates, "Life-Saver" badges, and community leaderboards. Partnerships with corporate sponsors could also provide

donors with tangible incentives, such as health insurance discounts or wellness rewards.

- **Expansion of Institutional Partnerships:** Strategic collaborations with International NGOs (like the Red Cross), private hospital chains, and university campus groups will be formalized. By onboarding more certified collection centers, Rakt Setu will ensure that users in rural or underserved areas have the same access to safe donation points as those in urban hubs.
- **Tele-Consultation and Post-Donation Care:** The platform plans to include features for post-donation health tracking, providing donors with access to medical advice and nutritional guidance to ensure their recovery is managed professionally, further encouraging repeat donations.

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APPENDIX

The complete source code of Rakt-Setu is available at - <https://github.com/praptisharma28/Rakt-Setu>

The key contents of the repository include:

- Client-Side: A React single-page application (SPA) providing a responsive user interface and native-like performance on mobile and desktop browsers.
- Server-Side: A Node.js and Express.js backend that manages business logic, JWT-based security, and real-time API communications.
- Data Layer: MongoDB schemas utilizing GeoJSON objects and 2dsphere indexing for high-speed, proximity-based donor matching.
- DevOps Tools: Docker configuration files (Dockerfile and docker-compose.yml) to ensure consistent containerized deployment across all environments.
- Third-Party Integrations: Implementation of Firebase Cloud Messaging (FCM) for instant notifications and the Google Maps API for donor navigation.

RESEARCH PAPER

The research paper titled "Rakt Setu – An Integrated Mobile Platform for Expediting Blood Donation and Request Management" (Paper ID: ICICT-092), has been successfully accepted and presented at the **9th International Conference on Inventive Computation Technologies (ICICT-2026)**, organized by Tribhuvan University, Nepal. This academic work provides a comprehensive examination of the platform's MERN-stack architecture, its core functionalities, and its containerized deployment strategy using Docker. Furthermore, the presentation highlighted critical empirical performance analyses, demonstrating how Rakt Setu optimizes emergency response through real-time geospatial coordination and high-speed notification delivery.

Authors: Ayush Srivastava, Shriyansh Srivastava, Shaurya Jain, Prapti Sharma

Supervisor: Mr. Thammali Gangadhar

Institution: KIET Group of Institutions, Ghaziabad, Delhi-NCR, UP, India

PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION

**9th International Conference on
Inventive Computation Technologies**
ICICT 2026
15-17, April 2026

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Acceptance Letter

Paper ID	Title	Author(s)
ICICT-092	An Integrated Mobile Platform for Expediting Blood Donation and Request Management	Ayush Srivastava, Shaurya Jain, Prapti Sharma, Shriyansh Srivastava, Thammali Gangadhar

Dear Author(s),

We are pleased to inform you that your paper has been accepted for presentation and publication at the 9th International Conference on Inventive Computation Technologies (ICICT 2026), to be held in Lalitpur, Nepal, during April 15–17, 2026. Your manuscript was reviewed by subject experts, and based on the reviewers' recommendations, it has been selected for inclusion in the conference proceedings. The accepted papers will be submitted for publication in IEEE Xplore, subject to IEEE compliance requirements. Authors are requested to complete the final submission and registration process within the prescribed deadlines.

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We congratulate you on your acceptance and look forward to your participation in ICICT 2026.

Best Regards,



Dr. Subarna Shakya
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Rakt Setu - An Integrated Mobile Platform for Expediting Blood Donation and Request Management

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Abstract— The critical shortage of timely blood supply in emergency medical situations is a significant global healthcare challenge. This challenge becomes more serious due to poor communication, logistical difficulties, and the absence of real-time coordination between donors and seekers. The primary objective of this research is to present a high-performance, integrated web application – Rakt Setu, that automates the end-to-end blood request management process through real-time geospatial coordination. Rakt Setu's architecture uses MERN (MongoDB, Express.js, React, Node.js) stack to guarantee a full-JavaScript, scalable, and maintainable codebase. The website has a dual-role architecture in which seekers create urgent, location-based requests to find suitable donors or blood banks. The smart geospatial matching tool, built into the platform and reliant on the MongoDB in-built geospatial query tool to alert nearby compatible donors immediately, is one of the innovations. While the Node.js backend is used to handle real-time APIs, the frontend is a single-page application (SPA) that is built on React. The entire application is bundled together with Docker, which provides a consistent environment of development and deployment. When one accepts a request, a secure, two-way, exchange of contact information is made, which guarantees direct communication. The application also connects to the Google Maps API to allow travelling and cooperates with certified hospitals to offer safe and authorized spots of donation. The long-term goal of this end-to-end workflow is to significantly lower the response time in emergencies and enhance donor participation as well as overall effectiveness of blood supply chain between request origination and assisted donation.

Keywords— Blood donation, MERN stack, Real-time Notification System, Geospatial Query, Docker containerization, Healthcare Logistics.

I. INTRODUCTION

BLOOD is an indispensable, life-saving resource in modern medicine, crucial for treating trauma victims, surgical patients, and individuals with chronic conditions. However, many regions face chronic shortages due to seasonal donor

scarcity, inefficient inventory management, and a lack of coordinated communication. Addressing this blood deficit is essential to help avoid delays in critical blood transfusions for people in need. The proliferation of smartphones presents a transformative opportunity to address these inefficiencies through dynamic, location-aware, and real-time coordination.

Annual Requirement in India

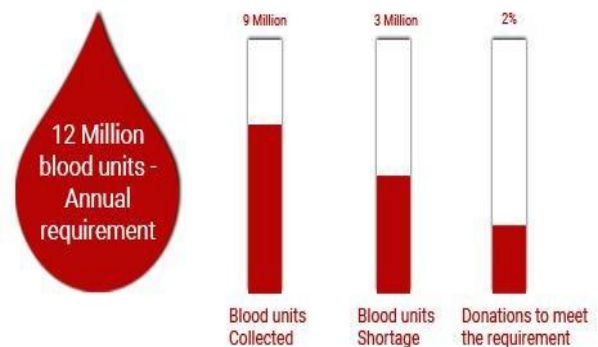


Fig. 1. Blood Shortage in India [1]

While various digital solutions have been introduced, a critical gap analysis of earlier studies reveals significant operational weaknesses. Early platforms such as "Friends2Support" [3] established large databases which remained fundamentally passive, requiring seekers to manually contact individual donors which is a time-consuming process and often fails during the "Golden Hour" of medical emergencies. Institutional portals like "E-Rakt Kosh" [5] have improved inter-bank coordination but do not enable direct donor-seeker matching for urgent needs. Existing mHealth apps, such as the Red Cross Blood Donor app [4], primarily function as one-way notification systems rather than interactive ecosystems. Furthermore, reliance on unregulated social media crowdsourcing does not have a

structured response mechanism and has privacy risks. This research proposes a holistic, closed-loop mobile ecosystem built on the MERN stack to bridge these identified gaps.

The primary contribution of this work is an integrated platform – “Rakt Setu”, which is built on the MERN stack that distinguishes itself through:

- A dynamic, proximity-based matching algorithm based on MongoDB's geospatial indexing.
- A secure, bidirectional communication protocol driven by a backend built with Node.js and Express.js.
- A seamless user experience is provided by a responsive React-based frontend.
- A containerized and globally deployed architecture using Docker, ensuring scalability and reliability.

This paper details the MERN-based architecture, functionality, and the Docker deployment of Rakt Setu.

II. LITERATURE REVIEW

Three primary areas of research have been conducted on the use of information technology to optimize blood supply chains: integrated mobile health (mHealth) apps, inventory and supply chain optimization, and donor management systems.

A. Donor Management and Recruitment Systems

A substantial amount of research focuses on methods for managing, recruiting, and retaining donors. In North India, Sachdev S, Mishra SK, Marwaha N, and Avasthi [6] carried out a study to determine college-bound students' attitudes about volunteer blood donation, including information gaps and inconvenient times. Early alternatives included digital platforms such as "Friends2Support" [3], which established online databases of voluntary donors. Although they are effective at creating vast databases, their main drawback is that they are passive; in an emergency, seekers must personally contact several donors from a list, which is a time-consuming and inefficient operation. They don't have automatic, intelligent matching or real-time availability checks.

B. Inventory and Supply Chain Management

There are various research works which focus on the logistics of blood inventory management after collection. For example, "E-Rakt Kosh" portal [5] of the Indian Government provides a centralized view of blood inventory across numerous banks, significantly improving inter-bank coordination. Osorio et al. [7] modelled and optimized the logistics of the blood supply chain using discrete-event simulation. It focused on reducing waste and improving distribution effectiveness. Although such platforms are important, they operate at the institutional level and do not help in quickly locating a donor for a specific, urgent need that is not covered by the current inventory.

C. Integrated mHealth Applications

The rise of smartphones led to the creation of mobile applications and websites for blood donation. The Red Cross has a site called Blood Donor [4] that assists the user in

locating blood drives and making appointments. It functions largely as a transmission system in terms of blood banks to donors. With a narrower target, the BloodLink Navigator System [8] was developed to enhance blood-request system and streamline the process of blood donation. This system offers an effective platform where the seekers, donors, and administrators can handle blood-related events effectively.

The challenges of adopting these systems have been researched. It has been discovered that a mistrust and the absence of perceived reliability in India makes it difficult to implement blood donation applications. They indicated that collaboration with reputable organizations, such as famous hospitals, could be used to deal with such problems. Also, some of the suggested systems such as Blood donor searching android application with GPS tracking by Sangeetha M., Dhivya A., Ramya R., and Sharmila N., [10] is intended to provide the ability to view the details of the nearby donors and easily connect with the donors in case of an emergency. This platform does not however publish the information of the donor to the seeker within 90 days hence impedes two way communication.

D. Social Media and Crowdsourcing

Social media platforms like Facebook and Twitter have become popular ways to request blood. They help a seeker to send a blood request to lakhs of people in one go. Facebook's "Blood Donations" feature lets users sign up as donors and get alerts about nearby needs. This method benefits from its large network. However, it is an unregulated, unreliable process that involves various privacy concerns and relies on the popularity of a post. Thus, social media platforms do not ensure a structured and guaranteed response towards a blood request.

E. Identified Research Gap

Previous research works and the existing platforms have made important contributions, yet they do not provide real-time donor matching and secure communication. Our proposed platform – Rakt Setu aims to fill this gap by creating a managed ecosystem that connects an urgent blood request with a successful donation. It works to connect a donor and a seeker directly without involving any third-party or middlemen.

Feature	Passive Directories (e.g., Friends2Support [3])	Institutional Portals (e.g., E-Rakt Kosh [5])	Social Media / Crowdsourcing	Proposed System (Rakt Setu)
Operational Model	Manual search from list	Institutional inventory	Unregulated broadcast	Real-time seeker-donor matching
Matching Engine	None (Static database)	Centralized stock view	Popularity/Algorithm-based	Automated Geospatial (2dsphere)
Communication	One-way/Seeker initiated	Inter-bank coordination	Unregulated/Private DMs.	Secure Bidirectional Exchange
Privacy Layer	High exposure of data.	Secure (Institutional)	Significant privacy concerns	Contact exchange only on acceptance
Logistics Support	None	General location mapping	None	Integrated Google Maps Navigation

Table. 1. Comparative Evaluation

III. RESEARCH METHODOLOGY

This research follows a structured Software Development Life Cycle (SDLC) Prototyping approach. Unlike traditional methods, this allows for the creation of a working model which can be refined until it meets all the necessary requirements of the web app.

A. The Four Research Stages

1. Requirement Analysis – A comprehensive analysis was performed to identify the reasons for failure of existing platforms and the essential requirements for an effective blood donation management platform. We reviewed previous studies and found that "communication latency" - the delay after which the donor receives a message, is the biggest hurdle in blood supply chains.

2. System Design & Data Modelling – The second phase of the research process is the drawing of blueprints of the app. A dual-role system has been implemented in which users are made different in case they are a Donor or Seeker. On MongoDB, we designed a certain data schema using geospatial indexing. This enables the database to consider the position of a user as a point on a map and not a piece of text. This enables milliseconds distance-based searching.

3. Implementation & Integration - This is the construction stage with MERN stack. We have also incorporated Google Maps API to guide the donors to donation points. We relied on the Firebase (FCM) to dispatch "Push Notifications" that appear on the phone of a donor immediately.

4. Empirical Testing – It is the final stage and here we measured the speed of the system. We attempted the time-consumption of the query to locate a donor with a radius of 15-20km (Query response time). We were also able to measure the time interval of clicking "Request" on a Seeker to the phone of the Donor receiving a message (Notification Latency).

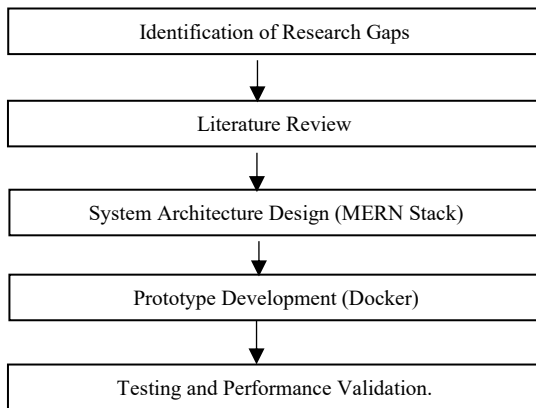


Fig. 2. Research Methodology Block Diagram

B. Model Parameters

Stage	Key Parameter	Purpose
Authentication	Auth Protocol (OTP)	To ensure user authenticity
Search Logic	2dsphere query	To find the nearby donors within a specified range

Notification	Multi cast push	To send one alert to many people at the same time
Donor Safety	90-day time interval	To ensure mandatory interval between two donations
Deployment	Docker containers	To ensure that the app runs on any system or server

Table. 2. Model Parameters

C. Procedural Workflow

Phase 1: User Registration and Data Collection

The first step is onboarding users on our platform, especially donors. Individuals can register themselves on the platform, selecting a specific role: Donor or Seeker. For building a comprehensive profile, the users provide essential details which include - personal information, contact information, their blood group, and the pin code (location data). Apart from individuals, blood banks can also register themselves on our platform by providing their contact details. The registration process is supported by OTP based user verification to build authenticity.

Phase 2: Request Initiation and Intelligent Matching

When a Seeker requires blood, they submit an urgent request through the platform. This request is immediately processed by the system's backend server, which identifies all registered donors as well as the blood banks who have the matching blood type, and are located within a predefined geographical radius of the request. The system then automatically sends a real-time push notification to these matched donors simultaneously. The Seeker also gets a list of Blood Banks along with their contact details, which have the requested blood type available in their stock. Once a seeker submits a request, he cannot make another request until and unless the previous request has been fulfilled. This assists in reducing the number of blood requests being made by one person at the same time.

Phase 3: Donor Response and Secure Communication

The donors that have their blood group matching that required are notified by the platform and the mobile devices and via the e-mails. The donors contacted can accept or reject the request depending on their location. In case the donor accepts the request, two-way sharing of contacts information (via direct SMS/ emails) takes place between the seeker and the donor. This move will not reveal any information about the donor or the recipient hence will not expose the information of the involved parties too early. The site also gives directions to the donor to the point where the seeker wants.

Phase 4: Donation Fulfilment and System Update

The last stage of the working process is real blood donation which is organized by hospitals/NGOs and updation of the database. Our platform provides the donor with the assistance of navigation using Google Maps and helps him reach the hospital, where the usual blood donation processes are conducted. After the donation process is done, the database of the platform is updated. The donor is then stipulated as not available to further donations temporarily over a given time period, in order to adhere to mandatory health intervals of donations. The donor dashboard has now shown the donation

that has been made and the seeker is permitted to make new blood requests.

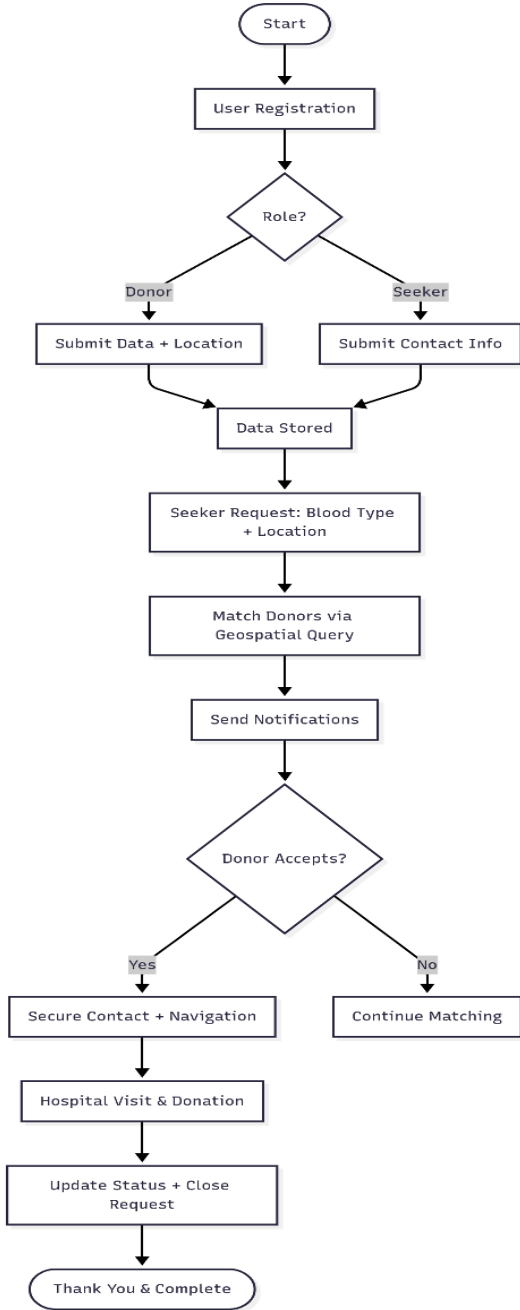


Fig. 3. Process Flowchart

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

Rakt Setu has been built on the MERN stack architecture, which is a cohesive full-stack JavaScript framework. It is widely used in building dynamic web applications due to its great efficiency, scalability, maintainability, and performance.

A. User Roles and Data Models

Our system has two primary user roles with tailored data models:

1 - Donors: Donors are the users who are willing to donate blood voluntarily. Detailed information such as name, age, gender, contact info, address, and medical data (blood

group, last donation date etc) is provided by the donors during registration. This data is stored securely and is essential for intelligent matching during urgent blood requests. (Table1.)

2 - Seekers: Seekers are those users who urgently need blood during emergencies. Seekers can register on our platform or use it as guests (for one time use) to generate blood requests. They too provide essential information such as the required blood type, the patient's location and their contact number, in order to submit a request.

Field	Description
donor_id	Unique donor ID
name	Full name
age	Age of donor
gender	Gender (M/F/Others)
blood_group	Blood group (A+/A-/B+/B-/O+/O-/AB+/AB-)
email	Email ID
contact_info	Phone number
address	Residential address
pincode	Pincode
password_hash	Encrypted passwords
fcmToken	Firebase token for notifications

Table. 3. Donor table

B. Technology Stack (MERN)

The system is implemented using MERN stack and the breakdown of technologies is discussed below:

1 - Frontend (React): The user interface of the web application is built using React. Multiple React libraries have been used to develop the frontend of the system. To illustrate, React router is an application routing the traffic between the various views (Registration page, Donor Dashboard, Seeker Dashboard, Blood Bank Page) of the webpage and Axios is employed in the smooth operation of HTTP communication with the backend API. The UI is responsive and has a native application-like behaviour on both the mobile and desktop browsers.

2 - Backend (Node.js + Express.js): Node.js and Express.js have been used to develop the backend logic of the web app. The backend layer is responsible for handling all HTTP requests from the frontend, implementing user authentication and authorization using JSON Web Tokens. It also executes the business logic, including the donor-matching algorithm and communicating with the database via Mongoose, an Object Data Modelling library for MongoDB.

3 - Database (MongoDB): MongoDB is a NoSQL document database. We use MongoDB for database implementation as it is known for its flexibility and native support for geospatial data. Here, the user locations are stored as GeoJSON objects, which allow for highly efficient proximity-based queries. A 2dsphere index is created on the location field to optimize these queries.

C. Deployment and DevOps: Docker & Vercel

We have used a modern DevOps approach to ensure consistency and scalability of the application:

Docker Containerization: The entire web-application is containerized using Docker. A Docker file defines the environment for the Node.js backend, and docker-compose.yml can be used to orchestrate the backend and MongoDB services together. This guarantees that the application runs identically across all environments—development, testing, and production—eliminating the "it works on my machine" problem.

Frontend Deployment: The React frontend is deployed on Vercel. Vercel provides:

- A Global CDN: The frontend assets are served from locations closest to the user worldwide, ensuring fast load times.
- Serverless Functions: The API routes can be implemented as Vercel Serverless Functions. However, currently we have the Express server maintained separately for WebSocket and real-time capabilities.
- Automatic CI/CD: Connecting the GitHub repository to Vercel triggers automatic deployments on every git push, streamlining the development workflow.

Backend Deployment: We can easily deploy the Node.js/Express backend API on platforms like, Render, Vercel or a cloud VM (e.g., AWS EC2) that supports running Docker containers. Environment variables are used to manage configuration settings, such as API keys securely across different deployment stages.

D. Core Workflow

The operational workflow is the core of the platform. All the processes follow a specific sequence of execution and are designed to ensure minimal latency. (Fig6.)

1 - Request Generation: The seeker submits a blood request through a structured form available in the Seeker dashboard. The frontend application then sends a request containing the JSON payload (blood type, location) to the /api/requests endpoint on the backend API server, initiating the emergency workflow to find the most suitable donors.

2 - Proximity Matching: The Express server receives the HTTP request and invokes the corresponding controller function. This function executes a geospatial query on the donor collection, to filter out available donors with a matching blood type available in the mentioned pin code or nearby areas which lie within acceptable radius, ensuring millisecond response times.

3 - Real-Time Notification: After successful execution of the geospatial query, the server retrieves the list of matching donors. A high-priority, multicast push notification is sent to all the devices associated with the donors' stored fcmTokens. The matched donors receive this notifications on the application as well as on their e-mails. This ensures immediate alerting without the need for the donor to have the application open.

4 - Donor Acceptance & Secure Contact Exchange: The donor can view all the blood requests on the donor dashboard. When a donor accepts a request, his/her acceptance is communicated from the React app to the backend via a PATCH request to the specific endpoint for that request. The server then validates the action and securely triggers a bidirectional exchange of contact information, revealing the seeker's phone number to the donor and vice-versa. The donor also gets the exact location of the seeker through the 'Navigation' option available on his dashboard, only upon this explicit confirmation.

5 - Navigation: Following the acceptance of a blood request by a donor, the React frontend dynamically renders an interactive map interface using the Google Maps JavaScript API, on the Donor dashboard. It automatically plots the route from the donor's current location to the designated hospital/seeker's location. This helps in donor's easy navigation and facilitates timely arrival.

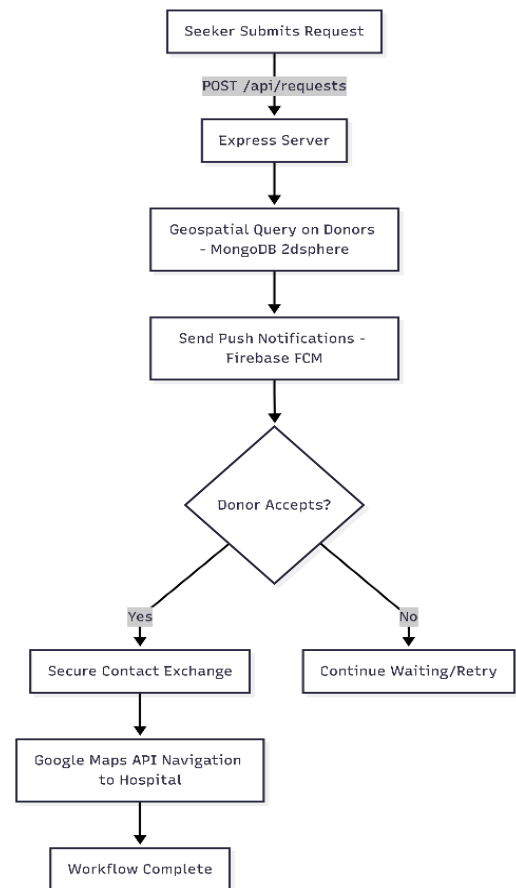


Fig. 4. Seeker Request and Acceptance

V. IMPLEMENTATION DETAILS

The whole platform has been implemented using the JavaScript ecosystem of the MERN stack. This allows for easy execution of the designed features and maintaining the efficiency of the system.

Frontend (React): The frontend has been built with React App for bootstrapping. The React Context API manages global state like user authentication. The frontend is designed as a Progressive Web App to offer app-like features such as offline functionality and home screen installation.

Backend (Node.js + Express): The backend is based on the use of Node.js and Express. The middleware employed by the server to provide security headers, cross-origin request to the frontend, express rate-limit to prevent API abuse, and elegant MongoDB object modelling.

Database (MongoDB Atlas): Database is deployed on MongoDB Atlas, the official cloud platform regarding MongoDB. This offers automatic backups, scaling and high availability without incurring the overhead of the manual database management.

Security: Hashing of passwords has been done using bcryptjs, which is of high standards. All API endpoints have authentication middleware. Secrets are stored in the environment variables and not the codebase.

VI. RESULTS AND DISCUSSIONS

The practical implementation of the platform along with the performance results obtained during system testing are discussed below. The system was evaluated to ensure that it works efficiently and responds quickly enough to meet the urgent needs of emergency medical situations.

A. System Implementation and Visual Workflow

Rakt Setu has been successfully implemented using the MERN stack architecture. Various modules of the web app - from user registration to the blood request fulfillment are shown below:

User Onboarding: The onboarding interface (shown in Figure 5) allows users to register themselves on the Rakt Setu platform as donors and seekers by providing essential demographic data and blood group details. The onboarding process is aided with OTP based authentication.

Fig. 5. User registration

Seeker Dashboard: Figure 6 shows the seeker dashboard which displays the real-time status of an active blood request

along with a list of blood banks having the stock of matching blood.

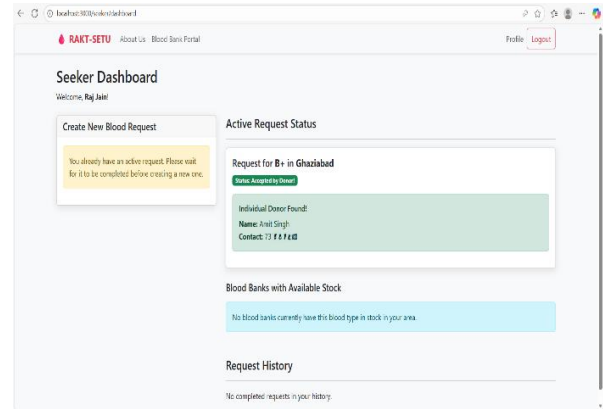


Fig. 6. Seeker Dashboard

Donor Dashboard: Figure 7 illustrates the donor dashboard, highlighting urgent local requests and providing an integrated navigation button for immediate response, once a request is accepted.

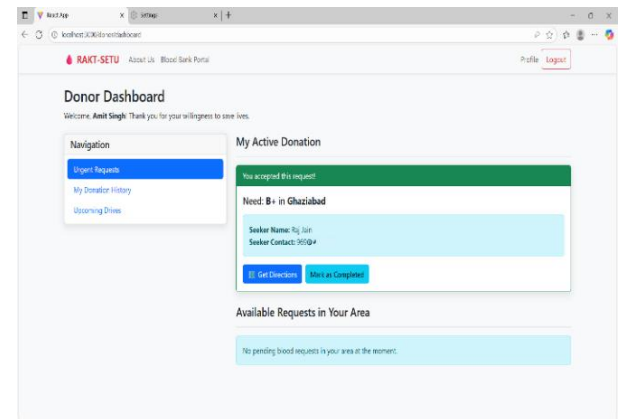


Fig. 7. Donor Dashboard

B. Empirical Performance Analysis

The system was tested for latency, delivery speed, and scalability to experimentally validate the research findings:

System Latency (Matching Speed):

We measured the time the Node.js backend takes to perform a geospatial query on the MongoDB database using the 2dsphere index. According to the testing results, the MongoDB 2dsphere index allows the backend to identify compatible donors within a 15km radius in less than 50ms.

Notification Delivery Time:

The speed of communication is critical for saving lives. We tracked the time from request submission to the donor receiving an alert via Firebase Cloud Messaging (FCM). Almost 95 out of every 100 alerts took less than 2 seconds to reach the donor devices and the 2nd email alerts took an average of 5 seconds.

Scalability and Reliability:

The system was able to manage a considerable amount of traffic because of Docker containerization. The application

also maintains the same volume of memory regardless of where it is running. The Node.js server was able to handle a high number of 500 parallel geospatial queries, without a major decline in the response rate.

VII. RECOMMENDATIONS AND FUTURE DIRECTIONS

This research is a solid basis of refining real-time blood donation and request management. It also reveals some strategies of improving the Rakt Setu platform to use it on a long-term basis. Next in line are the methods by which the participation of donors can be enhanced and the overall performance of the system can be enhanced.

A. Technical Improvements

1. Application of AI: Artificial Intelligence is applicable in finding donors who have higher chances to positively respond to a request depending on how they did respond in the past.

2. Application of Blockchain: Blood units can be tracked with the help of a digital ledger. This would contribute towards making supply chain transparent and safe.

3. Developing Native Apps: Android and iOS apps will be created to help in utilizing superior location tracking and push notifications. This would enhance the experience of the users.

B. Strategic Growth

1. Integration with Government Portals: The system can be directly connected to portals such as E-Rakt Kosh [5] to get more information about the inventories.

2. Gamification: It is possible to introduce a reward system that will motivate individuals to donate more.

VIII. CONCLUSION

Rakt Setu is a complete web-based application to support blood donation and management of requests. It deals with other pressing issues of blood seekers, when there is an emergency situation and an urgent need of blood. This piece of research has enhanced the existing technology in a much better way and simplified the process of donating as well as requesting blood online. MERN stack architecture offers good performance, scalability, and is an architecture that is not difficult to maintain. Containerization with Docker provides an opportunity to be deployed consistently across the board. One of the key findings in this study is the use of the 2dsphere indexing of the MongoDB to match donors based on their proximity. This is what allows real-time connections and this reduces the emergency response time by matching donors who are compatible within milliseconds. Secure two-way communication protocol is used to maintain the privacy of the user and to make sure that the contact information is shared upon a blood request being accepted by a donor. The web application is also linked with Google Maps API that would allow smooth navigation to confirmed healthcare facilities and secure donor safety and reliability of the procedure. The

partnerships with healthcare institutions and NGOs are some of the strategic partnerships that make blood donation activities reliable. Rakt Setu is an enormous enhancement of the old platforms. In comparison with the old platforms that operated in isolation and executed few functions, Rakt Setu develops a framework that coordinates the entire operation including urgent request to successful donation. It unites the seekers, donors and the blood banks at one platform and all are interconnected where they can communicate directly without risk of privacy. This approach may alter the response to blood emergencies in communities particularly those that handle severe blood shortages.

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